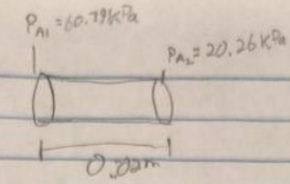
P_{A_1} = Partial Pressure of Methane at Point 1
P_{A_2} = Partial Pressure of methane at Point 2

Josh Schroeder
CHE 313
HW 2

75/100



18.1-1 $P = 101.32 kPa$

20/20 $T = 298 K$

$$J_{AB} = -D_{AB} \frac{dC_A}{dz}$$

$$J_{AB}\, dz = -D_{AB}\, dC_A$$

$$J_{AB} \int_1^2 dz = -D_{AB} \int_1^2 dC_A$$

$$J_{AB}(z_2 - z_1) = -D_{AB}(C_{A_2} - C_{A_1})$$

$$J_{AB} = \frac{-D_{AB}(C_{A_2} - C_{A_1})}{(z_2 - z_1)}$$

$PV = nRT$

$$\frac{P}{RT} = \frac{n}{V} = C$$

$$C_{A_2} = \frac{P_{A_2}}{RT}$$

$$C_{A_1} = \frac{P_{A_1}}{RT}$$

$$\longrightarrow J_{AB} = \frac{-D_{AB}(P_{A_2} - P_{A_1})}{RT(z_2 - z_1)}$$

D_{AB} for He-CH_4 is 0.675×10^{-4}

$$J_{AB} = \frac{-(0.675 \times 10^{-4} m^2/s)(20.26 kPa - 60.79 kPa)\left(\frac{1000 Pa}{1 kPa}\right)}{\left(8314.3 \frac{m^3 Pa}{kg mol \cdot K}\right)(298K)(0.02m)} = \boxed{5.52 \times 10^{-5} \frac{kg\, mol\, A}{s \cdot m^2}}$$

18.2-1 For a mixture of Ethanol(g) (CH_3CH_2OH) and methanol(g) (CH_3OH)

20/20

Fuller et Al

$$D_{AB} = \frac{1.00 \times 10^{-7}\, T^{1.75} \left(\frac{1}{M_A} + \frac{1}{M_B}\right)^{1/2}}{P[(\Sigma v_A)^{1/3} + (\Sigma v_B)^{1/3}]^2}$$

For Ethanol (A) $\Sigma v_A = 2(16.5) + 6(1.98) + (5.48)1 = 50.36$

For Methanol (B) $\Sigma v_B = 1(16.5) + 4(1.98) + (5.48)1 = 29.90$

a) $P = 1.0132 \times 10^5 Pa = 1 atm$ at T_1

$T_1 = 298 K$

$T_2 = 373 K$

$$D_{AB} = \frac{1.00 \times 10^{-7}(298K)^{1.75}\left(\frac{1}{46.07 \frac{g}{mol}} + \frac{1}{32.03 \frac{g}{mol}}\right)}{(1\, atm)[50.36^{1/3} + 29.90^{1/3}]^2}$$

$$\boxed{= 1.064 \times 10^{-5} \frac{m^2}{s}}$$

T_2

$$D_{AB} = \frac{(1.00 \times 10^{-7})(373)^{1.75}\left(\frac{1}{46.07 \frac{g}{mol}} + \frac{1}{32.03 \frac{g}{mol}}\right)}{(1 atm)[(50.36)^{1/3} + (29.90^{1/3})]^2}$$

$$\boxed{= 1.58 \times 10^{-5} \frac{m^2}{s}}$$

b) Same but with

$P = 2.0265 \times 10^5 Pa = 2 atm$

$T = 298K$

$$\boxed{D_{AB} = 5.32 \times 10^{-6} \frac{m^2}{s}}$$

18.2-5 Dilute NaOH

a) at 25°C estimate Diffusivity

$$D_{Na^+} = 2.662\times10^{-7}\,\frac{\lambda_{Na^+}}{n_{Na^+}} = 2.662\times10^{-7}\left(\frac{50.1}{1}\right) = \boxed{1.334\times10^{-5}\,\frac{cm^2}{s}}$$

$$D_{OH^-} = 2.662\times10^{-7}\,\frac{\lambda_{OH^-}}{n_{OH^-}} = 2.662\times10^{-7}\left(\frac{197.6}{1}\right) = \boxed{5.26\times10^{-5}\,\frac{cm^2}{s}}$$

$$D_{NaOH} = \frac{n_{Na^+} + n_{OH^-}}{n_{Na^+}/D_{Na^+} + n_{OH^-}/D_{OH^-}} = \frac{1+1}{1/1.334\times10^{-5} + 1/5.26\times10^{-5}} = 2.13\times10^{-5}\,cm^2/s$$

b) at 15°C = 288 K

$$D_{AB} = 8.928\times10^{-10}\,T\,\frac{(1/n_{Na^+} + 1/n_{OH^-})}{(1/\lambda_{Na^+} + 1/\lambda_{OH^-})} = 1.61\times10^{-5}\,cm^2/s$$ 20/20

18.4-1 ? Not in book

Goes from 18.2-10 to 18.4-3 0/20

18.4-5 $N_A = -CD_{AB}\frac{dx_A}{dz} + x_A(N_A + N_B)$

$$N_A - x_A(N_A+N_B) = -CD_{AB}\frac{dx_A}{dz}$$

$$\int_{z_1}^{z_2}\frac{dz}{CD_{AB}} = \int_{x_{A1}}^{x_{A2}} -\frac{dx_A}{N_A - x_A(N_A+N_B)}$$

mention all the steps of integral

$$\frac{z_2-z_1}{CD_{AB}} = \ln\frac{(N_A - x_{A_1}(N_A+N_B))}{(N_A - x_{A_2}(N_A+N_B))}$$

15/20

$$1 = \frac{N_A}{N_A}\left(\frac{1}{N_A+N_B}\right)\frac{CD_{AB}}{z_2-z_1}\ln\frac{\left(\frac{N_A}{N_A+N_B} - x_{A_2}\right)}{\left(\frac{N_A}{N_A+N_B} - x_{A1}\right)}$$

$C = \frac{n}{V} = \frac{P}{RT}$

$x_{A_2} = \frac{P_{A_2}}{P}$

$$\boxed{N_A = \frac{2D_{AB}P}{RT(z_2-z_1)}\ln\left(\frac{1-\frac{P_{A2}}{2P}}{1-\frac{P_{A1}}{2P}}\right)}$$